

👉 XAI: Interactive Explanations — XAL and TalkToModel

FAMAF - UNC

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Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

Explainable Active Learning (XAL): Toward AI Explanations as Interfaces for Machine Teachers

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The wide adoption of Machine Learning (ML) technologies has created a growing demand for people who can train ML models. Some advocated the term “machine teacher” to refer to the role of people who inject domain

(Ghai et al. 2021)

🍷 Motivation: Machine Learning Bottleneck ---

Current ML development processes are bottlenecked by:

- **Machine learning is used** in a wide variety of fields
- **Resources and expertise** — labeling is costly
- **Asynchronous interaction** between Subject Matter Expert (SME), developer and the model
- Many AL paradigms have been developed; very little attention given to **improving human interaction** with AL algorithms

Efforts are underway to make machine learning more accessible.

🍷 Related Work: Active Learning ---

Definition

Active Learning (AL): a paradigm in which the model **queries** a human annotator for labels on the data points where it is most uncertain.



- Human annotators label data → model learns → model queries uncertain points
- AL tries to maximize model performance while minimizing annotation effort via **uncertainty sampling**.

🍷 Related Work: Interactive ML ---

- AL is sometimes considered within **interactive machine learning (iML)**
- iML approaches **value transparency over performance**
- Empirical studies demonstrate iML techniques lower need for data — but little else
- iML approaches are esoteric; **explanations as interfaces** could help non-ML experts

🍲 Problem Statement ---

Active Learning interfaces remain minimal and opaque

- But annotators get **no feedback** on why the model is uncertain
- Annotator cannot monitor **training progress**
- Annotator is unaware of the **effectiveness** of their teaching

Explanations show potential for better active learning.

Big Question: Can explanations improve the active learning process?

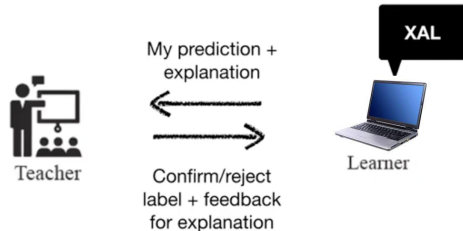
🍷 Summary of Contributions ---

- ➊ Propose a novel paradigm: **Explainable Active Learning (XAL)**
- ➋ Conduct an **empirical study** to investigate the impact of explanations on annotation experience

🍷 Contribution #1: The XAL Paradigm ---

Explainable Active Learning

- 1 Model sends prediction + **local explanation** to annotator
- 2 Annotator confirms or rejects the prediction
- 3 Annotator **provides feedback** on the explanation
- 4 Model updates from annotation **and** feedback



Mimics how humans teach and learn — explanations make it easier to **reject incorrect predictions**.

🍷 The XAL Paradigm: Key points ---

- Uses the explanation to answer “why am I giving this instance this prediction?”
- Explains specific instances, not global — better for non-experts.
- Improves labeling quality by exposing the model’s reasoning.

🍷 Contribution #2: Empirical Study ---

Dataset: Adult Income

- Binary prediction: annual income $>$ or $<$ \$80k
- Model: logistic regression with L2 regularization

Annotators receive a **customer profile** and must judge the income level.

Attributes	Values
Age	39
Workclass	Private
Years of Education	13
Marital status	Never married
Occupation	Executive & managerial
Race	White
Gender	Male
Hours per week	50

Customer profile

Experimental Design ---

- **Participants:** Amazon Mechanical Turk - 36 participants
- **Model:** Two learning stages (each participant completed both)
 - ▶ Early
 - ▶ Late
- **Three conditions:** (each participant assigned to one)

AL

Standard Active Learning —
customer profile only

CL

Coactive Learning — profile +
model prediction

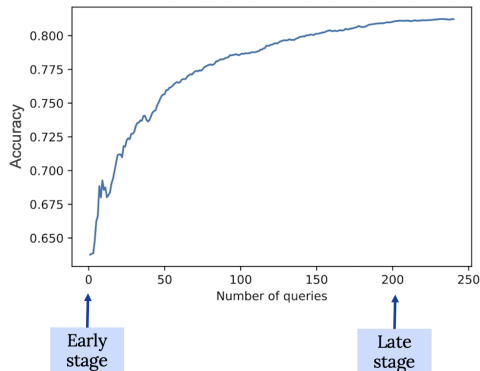
XAL

Profile + prediction +
explanation

- 20 annotations per experiment (condition + stage)
- Domain knowledge training (statistics, practice trials, \$2 bonus for consistency with ground-truth)

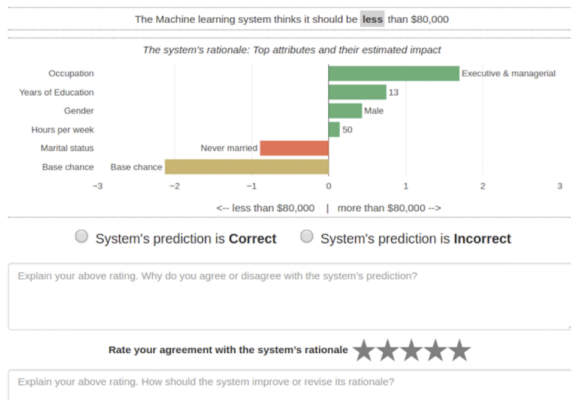
🍷 Design Choices: Active Learning Simulation ---

- Simulation used to define **early-stage** and **late-stage** models
 - ▶ Early stage → 0 queries (low accuracy)
 - ▶ Late stage → 200 queries (accuracy plateaus)
- Queried instances annotated using ground-truth labels in simulation



🍷 Design Choices: Explanation Format ---

Prediction
Explanation
Feedback



- Local feature importance bar chart (**top 5 features**)
- **Green** = positive influence on prediction
- **Red** = negative influence
- Orange baseline = model intercept (**base chance**)
- Features sorted by importance magnitude

🍷 Research Question 1 ---

How do local explanations impact the **annotation and training outcomes** of AL?

🍲 RQ1: Annotation + Learning Outcomes ---

Active (AL)
Coactive (CL)
Explainable (XAL)

Stage	Condition	Acc.	Acc. improve	F1	F1 im- prove	%Agree	Human Acc.
Early	AL	67.0%	13.7%	0.490	0.104	55.0%	66.7%
	CL	64.2%	11.7%	0.484	0.105	58.3%	62.1%
	XAL	64.0%	11.8%	0.475	0.093	62.9%	63.3%
Late	AL	80.4%	0.1%	0.589	0.005	47.9%	54.2%
	CL	80.8%	0.2%	0.587	0.007	55.8%	58.8%
	XAL	80.3%	-0.2%	0.585	-0.001	60.0%	55.0%

Compare to initial model performance (before 20 queries)

Key Finding

Overall, **human accuracy** and **% agreement** *decrease* in the later stage, while **model accuracy** increases.

Research Question 2 ---

How do local explanations impact **annotator experiences**?

H1

Explanations support **trust calibration**

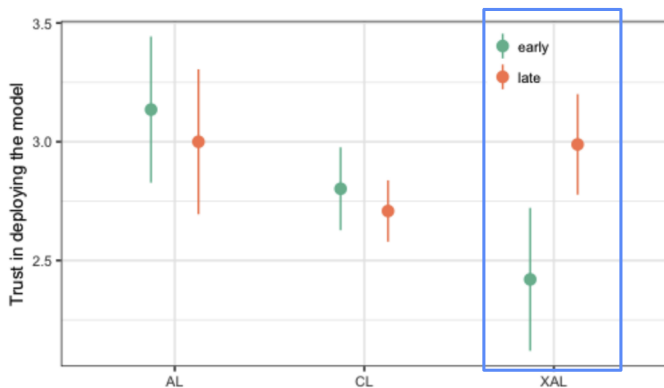
H2

Explanations improve **annotator satisfaction**

H3

Explanations increase perceived **cognitive workload**

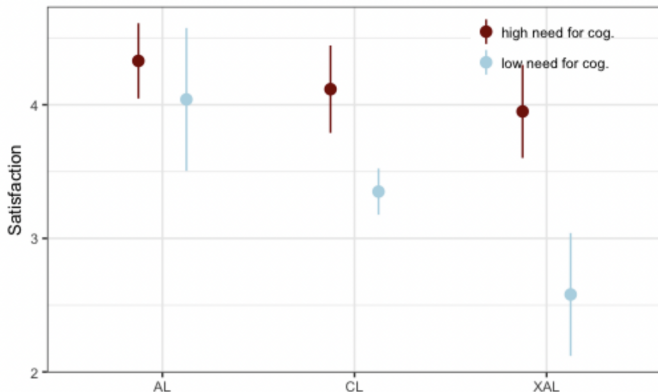
🍷 RQ2: Trust Calibration (H1) ---



H1 Supported

XAL users show **low trust early** (model is wrong) and **increasing trust late** (model improves). AL/CL users trust blindly throughout.

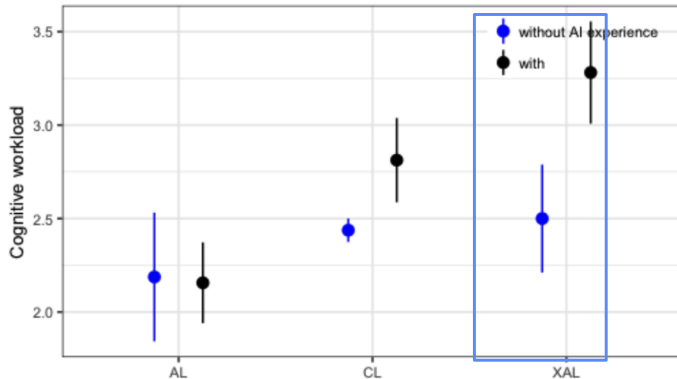
🍲 RQ2: Satisfaction (H2) ---



H2 Not Supported

Explanations have a **significant negative effect on satisfaction** for those with **low need for cognition** — they find explanations burdensome rather than helpful.

🍷 RQ2: Cognitive Workload (H3) ---



H3 Supported

XAL induces **significantly higher workload** for those **with AI experience** — they engage more deeply with the explanations.

Research Question 3 ---

How do **individual factors** impact annotation and annotator experiences with XAL?

H4

Annotators with **lower task knowledge** benefit more from XAL

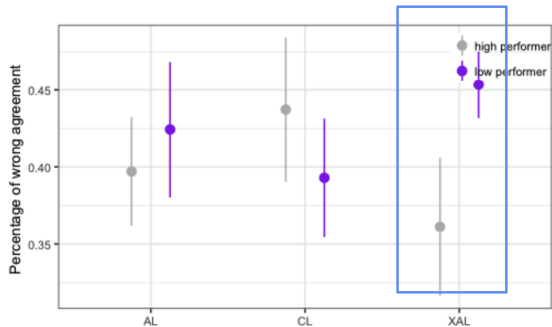
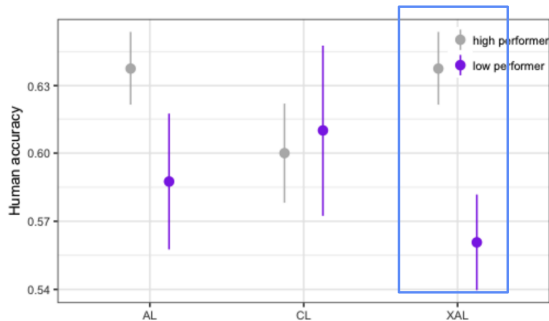
H5

Annotators **inexperienced with AI** benefit more from XAL

H6

Annotators with **lower need for cognition** have a less positive experience with XAL

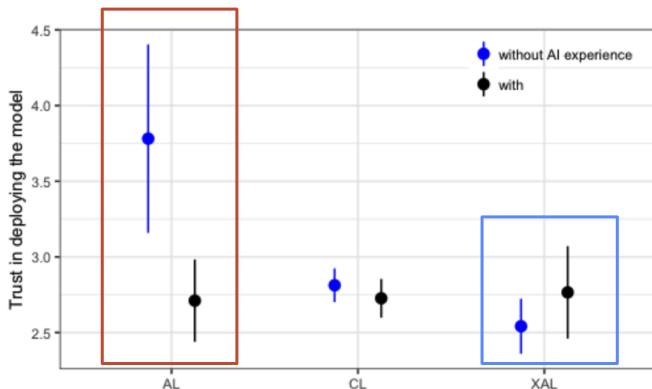
🍷 RQ3: Task Knowledge (H4) ---



H4 Not Supported

Participants with **less task knowledge** had **lower accuracies** and **higher blind trust** in XAL condition — they did not benefit more; they anchored on explanations.

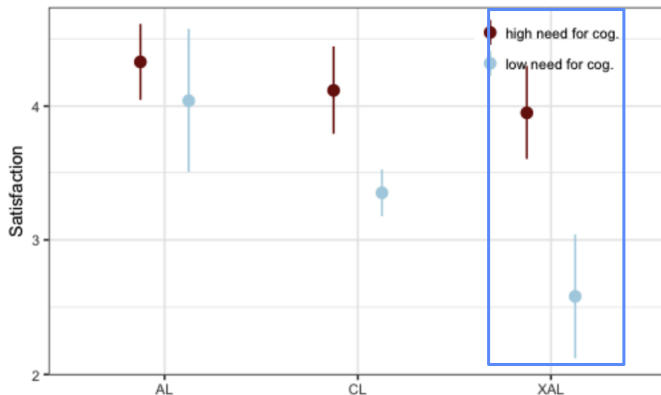
🍷 RQ3: AI Experience (H5) ---



H5 Supported

Explanations helped **calibrate trust for those without AI experience**. Users with AI experience showed higher trust regardless of condition.

🍲 RQ3: Satisfaction and Need for Cognition (H6) ---



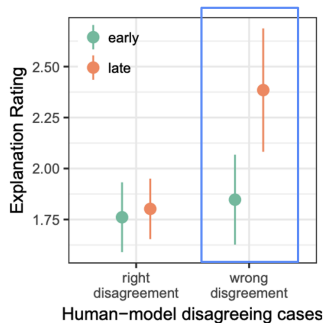
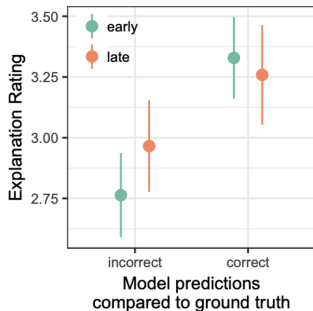
H6 Supported

Explanations have a **significant negative effect on satisfaction** for those with **low need for cognition** — they find explanations burdensome rather than helpful.

Research Question 4 ---

What kind of **feedback** do annotators naturally want to provide upon seeing local explanations?

🍷 RQ4: Explanation Ratings ---



Explainable (XAL) condition

Key Finding

Explanation ratings are **higher** when the model is **correct** and when annotators **wrongly disagree** with the model in later stage tasks — users judge explanations by outcome, not reasoning quality.

RQ4: Open Form Feedback ---

What participants wanted to tell the model:

- **Tuning feature weights** (81 mentions)
- **Removing, changing direction, or adding features** (28)
- **Ranking or comparing multiple feature weights** (12)
- **Reasoning about combinations and relations** (10)
- Logic to make decisions based on feature importance (6)

XAL Conclusions ---

Key Contributions

- New approach for machine teaching: **XAI + AL = XAL**
- XAL improves **trust calibration** and increases **cognitive engagement**
- Explanations enable annotators to act as **machine teachers**

Limitations

- Small N (36 AMT workers)
- Only uncertainty sampling
- Only one explanation method
- Only interpretable model (LR)
- What about image data (pixels)?

XAL Discussion Questions ---

- 1 Has anyone used AL in the past? Would adding explanations have been helpful?
- 2 What applications of XAL are you most excited about? In which scenarios would it work especially well or poorly?
- 3 Do you think XAL is feasible with less interpretable models or other explanation methods?

Paper 2: TalkToModel ---

Explaining Machine Learning Models with Interactive Natural Language Conversations Using TalkToModel

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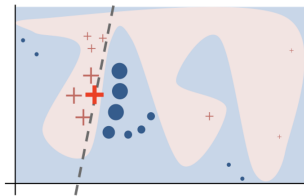
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Abstract

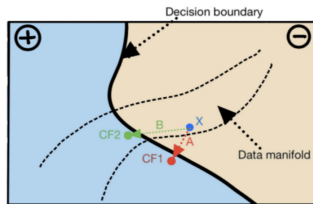
Practitioners increasingly use machine learning (ML) models, yet they have become more complex and harder to understand. To address this issue, researchers have proposed techniques to

(Slack et al. 2023)

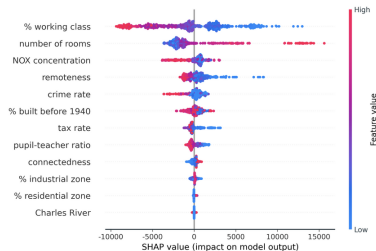
Motivation: The XAI Landscape ---



LIME Local linear approximation around a point



Counterfactuals Minimal changes to flip prediction



SHAP Feature importance via Shapley values

Motivation: The Practitioner's Dilemma ---

A nurse or doctor (or someone who is not an AI specialist) trying to use XAI:

- ➊ Which method to choose for my application?
- ➋ Which one would you trust?
- ➌ How should they interpret the explanations?
- ➍ How would they use these methods?
- ➎ How would they **interact** with the method?
- ➏ What if they have follow-up questions?

Motivation ---

Root problem

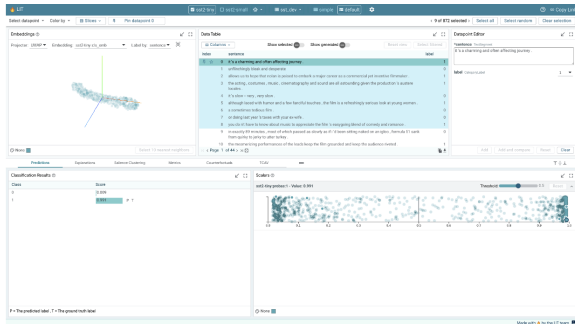
Every XAI tool requires its own expertise — there is no **universal, conversational interface**.

The XAI Bottleneck

- Simple explanations are a **bottleneck to adoption**
- There are inherently interpretable models; however, black box models are more flexible and accurate.
- Post-hoc methods are hard to use empirically:
 - ▶ Which method to pick?
 - ▶ How to interpret results?
 - ▶ How to ask follow-up questions?

🍷 Related Work ---

- **Language-Interpretability Tool (LiT):** Open-source NLP model understanding. Uses LIME, aggregate stats, counterfactuals.
- **What-If Tool:** Helps users perform counterfactual analysis for models.
- **explainerdashboard:** Used as baseline. Tab-based dashboard with SHAP, CFE, feature dependence.



Common problem

Relatively **high barrier to entry** and **no follow-up questions** — users are passive consumers of explanations.

Problem Statement ---

Definition

Design a system that makes it easy for **lay practitioners** to apply post-hoc interpretability methods to black-box models.

Desiderata:

- **Dialogue system** handling many conversation topics (data trends, specific predictions, etc.)
- **Variety** of data types and model classes (treatment prediction, risk of relapse, etc.)
- Does **not require** high ML expertise

Summary of Contributions: TalkToModel ---

TalkToModel

Open-ended dialogue for understanding any **dataset + classifier** pair

Capabilities: why a prediction occurred, how it changes if data changes, how to flip predictions, general data statistics, etc.

Problem

- ① Which method to choose?
- ② Which would a practitioner trust?
- ③ How to interpret?
- ④ How to use?
- ⑤ How to interact?

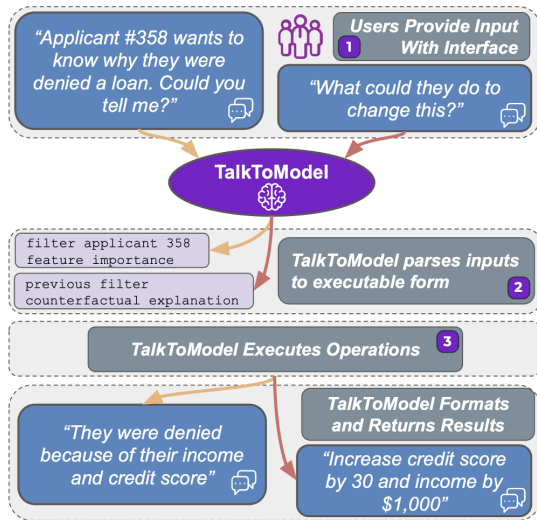
TalkToModel

- ① Uses **many** post-hoc methods
- ② Picks the **"best"** explanation
- ③ Answers in **natural language**
- ④ Only needs **model + data**
- ⑤ Communicate via **natural language**

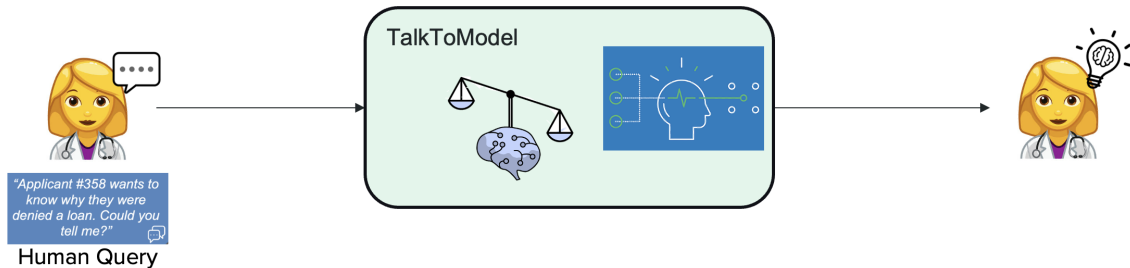
🍷 Talking to the Model: TalkToModel's Answers ---

Three components:

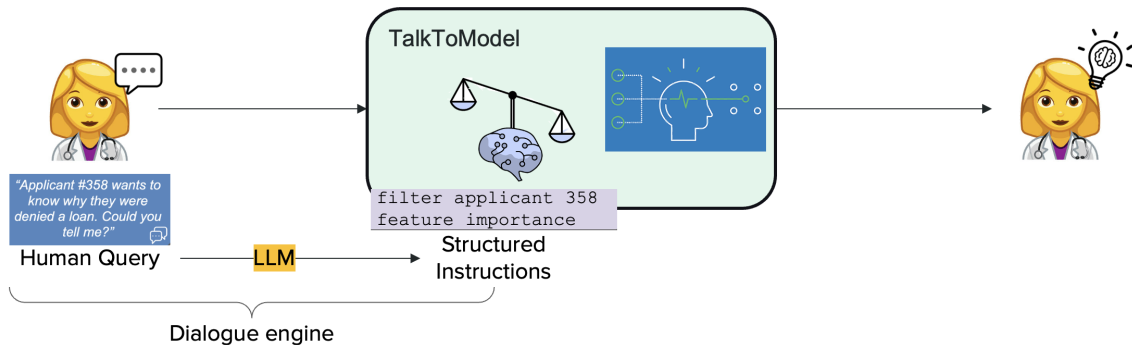
- 1 **Dialogue engine** — LLM backend translating utterances to operations
- 2 **Execution engine** — runs many explanations and picks the best one
- 3 **Text interface** — enables natural language conversations



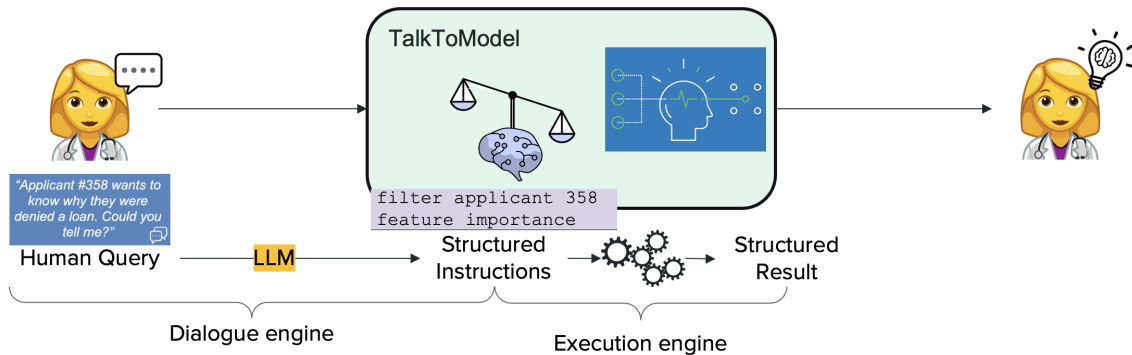
🍷 Talking to the Model: TalkToModel's Answers ---



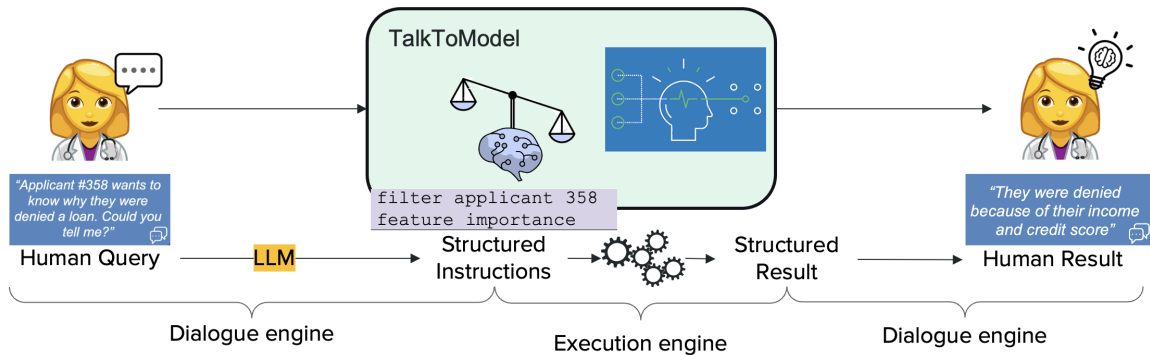
🍷 Talking to the Model: TalkToModel's Answers ---



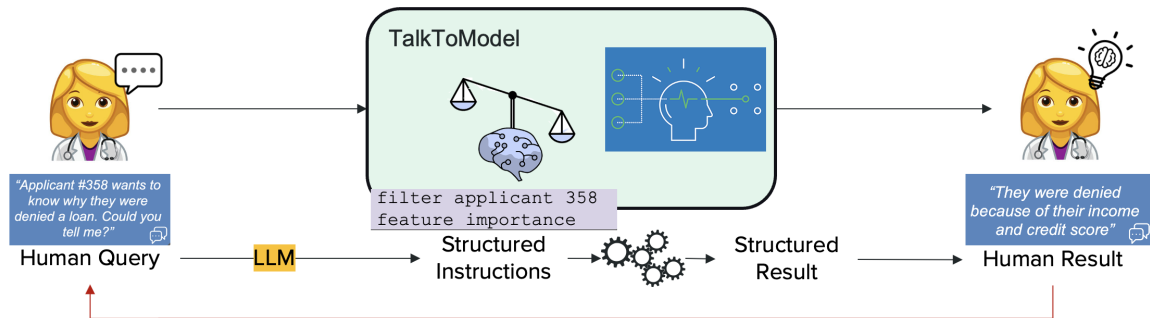
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🍷 Talking to the Model: TalkToModel's Answers ---



Text Interface - The result is used as context for the new question.

🍷 Dialogue Engine: The Grammar ---

Domain-Specific Language (DSL)

To represent the intentions behind user utterances in a structured form, TalkToModel relies on a grammar defining a domain specific language for model understanding.

The grammar includes:

- 1 All **operations** TalkToModel can run
- 2 The **arguments** for each operation
- 3 The **relations** between operations

How should instance A
change to get a different
prediction Q?



```
cfe(filter(test_data, id, A, =), 10, Q)
```



Dialogue Engine: Grammar Operations ---

	operation, arguments, and description
Data	<code>filter(dataset, feature, value, comparison)</code>: filters <code>dataset</code> by using value and comparison operator <code>change(dataset, feature, value, variation)</code>: Changes <code>dataset</code> by increasing, decreasing, or setting feature by <code>value</code> <code>show(list)</code>: Shows items in list in the conversation <code>statistic(dataset, metric, feature)</code>: Computes summary statistic for <code>feature</code> <code>count(list)</code>: Length of list <code>and(op1, op2)</code>: Logical “and” of two operations <code>or(op1, op2)</code>: Logical “or” of two operations
Explainability	<code>explain(dataset, method, class=predicted)</code>: Feature importances on <code>dataset</code> <code>cfe(dataset, number, class=opposite)</code>: Gets <code>number</code> counterfactual explanations <code>topk(dataset, k)</code>: Top <code>k</code> most important features <code>important(dataset, feature)</code>: Importance ranking of <code>feature</code> <code>interaction(dataset)</code>: Interaction effects between features <code>mistakes(dataset)</code>: Patterns in the model’s errors on <code>dataset</code>
ML	<code>predict(dataset)</code>: Model predictions on <code>dataset</code> <code>likelihood(dataset)</code>: Prediction probabilities on <code>dataset</code> <code>incorrect(dataset)</code>: Incorrect predictions <code>score(dataset, metric)</code>: Scores the model with <code>metric</code>
Conv.	<code>prev_filter(conversation)</code>: Gets last filters <code>prev_operation(conversation)</code>: Gets last non-filtering operations <code>followup(conversation)</code>: Respond to system followups
Description	<code>function()</code>: Overview of the system’s capabilities <code>data(dataset)</code>: Summary of dataset <code>model()</code>: Description of <code>model</code> <code>define(term)</code>: Defines <code>term</code>

Data: Supports data exploration and filtering operations.

Explainability: Supports interpretation and explanation of model behavior.

ML: Supports prediction and evaluation tasks for machine learning models.

Conversational: Supports contextual and follow-up interactions.

Description: Provides general information about functions, data, and models.

🍲 Dialogue Engine: Dataset-Specific Grammar ---

Challenge

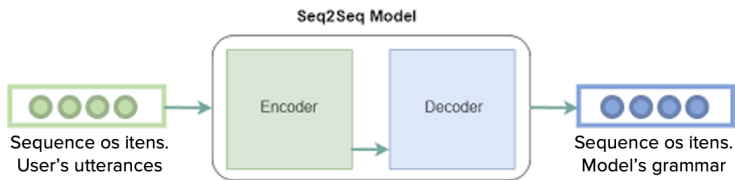
It is challenging to use a **general grammar** that works for all datasets.

TalkToModel uses a grammar that is **dependent on the dataset features**.

Example

The arguments for operations like **filter** or **topk** are automatically filled with the actual feature names from the user's provided dataset.

🍷 Dialogue Engine: Fine-Tuning LLM (seq2seq) ---



"To parse user utterances into the grammar, we fine-tune an LLM to translate utterances into the grammar in a **seq2seq fashion**."

Input: User utterances

"How likely is a 40-year-old woman to have diabetes?"

Output: Grammar parses

`likelihood(filter(data, age, 40, =))`

🍷 Dialogue Engine: Training Data Generation ---

- 1 Authors write **50 (utterance, parse) pairs** per domain — every operation appears $\geq 2\times$
- 2 MTurk: paraphrase each utterance **8 ways** = 400 pairs
- 3 MTurk: rate fidelity of paraphrase (keep $\geq 3/4$ avg over 5 raters)
- 4 Manual filtering by authors
- 5 Enumerate wildcards with dataset features $\rightarrow$ **20k–40k training pairs**

Dialogue Engine: Example Paraphrases ---

Original utterance

"What is your reasoning for determining if people older than 20 are likely to commit crimes?"

MTurk paraphrases

- "Why do you think people over the age of twenty are likely to commit a crime?"
- "How did you determine the likelihood of people over 20 committing crimes?"
- "Can you reason why people over twenty would likely commit crimes?"

🍷 Dialogue Engine: Responding Conversationally ---

Template-based responses

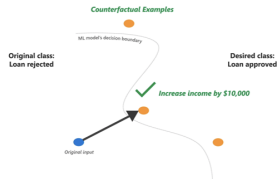
After TalkToModel executes a parse, it **composes the results** of the operations into a natural language response using **templates** associated with each operation.

- Each operation has an associated **response template**
- TalkToModel can run **multiple operations simultaneously** — it joins response templates ensuring semantic coherence

Dialogue Engine: Complete Pipeline ---

- ➊ **Constructing a grammar** — DSL with operations, arguments, relations
- ➋ **Generate fine-tuning data** — 50 pairs → wildcard enumeration → 20k–40k pairs
- ➌ **Fine-tuning LLM (T5)** — translate utterances to parses (seq2seq)
- ➍ **Respond conversationally** — templates composed into natural language

🍷 Execution Engine ---



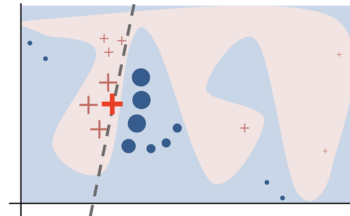
Counterfactual Explanations:

TalkToModel uses DiCE because it provides a diverse set of counterfactuals.



Data and Predictions Exploration

TalkToModel allow users to analyse predictions, inspect the model for errors, and analyse de data itself.



Post-hoc Feature Explanations

TalkToModel uses LIME and SHAP to make feature importance based explanations.

🍲 Execution Engine: Selecting the Best Explanation ---

Key design choice

Instead of providing raw LIME or SHAP values, TalkToModel **tests multiple methods** and reports the **most faithful** one.

Setup:

- f : model outputting probability y of a class
- ϕ : feature importances for model f on instance x (greater magnitude = higher importance)
- Methods evaluated: LIME with kernels $\{0.25, 0.5, 0.75, 1.0\}$ + KernelSHAP

🍲 Execution Engine: The Fudge Score ---

Intuition: More important features should cause **larger perturbations** in the prediction when noise is added.

$$\text{Fudge}(f, \mathbf{x}, \mathbf{m}) = \frac{1}{N} \sum_{n=1}^N |f(\mathbf{x}) - f(\mathbf{x} + \epsilon_n \odot \mathbf{m})|$$

Average magnitude of prediction perturbation when adding Gaussian noise **masked by m**.

🍷 Execution Engine: Faith Score ---

Feature Importance Faithfulness (Faith):

$$\text{Faith}(\phi, f, x, K) = \sum_{k=1}^K \text{Fudge}(f, x, \mathbf{1}(k, \phi))$$

where $\mathbf{1}(k, \phi)$ is a binary mask that selects the top- k most important features

- Computed for LIME (4 kernels) and KernelSHAP
- **Report the one with highest Faith**

Outcome

TalkToModel automatically selects the most faithful explanation — the user never has to choose.

Experiments: Overview ---

Experiment 1

LLM Experiment — Is the LLM accurately interpreting user questions?

Experiment 2

Grammar Experiment — Is the grammar expressive enough for all XAI questions?

Experiment 3

User Experiment — Do users prefer TalkToModel over a standard dashboard?

Experiment 1: LLM Evaluation ---

Goal: Is the LLM accurately interpreting user questions?

Method

- Create a **"Gold Dataset"**: ground-truth (utterance, parse) pairs specific to each domain
- Evaluate **Exact Match Accuracy** of LLM translation
- Compare along **easy** (IID) and **hard** (compositional) splits
- Compare few-shot GPT-J vs. fine-tuned T5 at different sizes

Experiment 1: Datasets and Gold Data ---

Domains

- **Diabetes**: 768 women, 8 features, 400→190 questions
- **Credit**: 1000 applicants, 20 features, 400→200 questions
- **Recidivism (COMPAS)**: 11757 defendants, 43 features, 400→146 questions

Data Collection

- 1 Authors: 50 (utterance, parse) pairs per domain
- 2 MTurk: 8 paraphrases per utterance
- 3 MTurk: fidelity rating (keep $\geq 3/4$ avg)
- 4 Manual filtering

Splits:

IID (Easy)

Order of operations **seen in training** — only arguments differ

Compositional (Hard)

Order of operations **not seen before** in training data

🍷 Experiment 1: LLM Results ---

	German			Compas			Diabetes		
	IID	Comp.	Overall	IID	Comp.	Overall	IID	Comp.	Overall
Nearest Neighbors	26.2	0.0	16.5	27.4	0.0	21.9	10.9	0.0	8.4
GPT-Neo 1.3B									
10-SHOT	41.3	4.1	27.5	35.9	0.0	28.8	40.1	7.0	32.6
20-SHOT	39.7	0.0	25.0	39.3	0.0	31.5	42.9	2.3	33.7
30-SHOT	42.9	0.0	27.0	39.3	0.0	31.5	41.5	4.7	33.2
GPT-Neo 2.7B									
5-SHOT	38.1	4.1	25.5	35.9	3.4	29.5	46.9	7.0	37.9
10-SHOT	38.1	6.8	26.5	40.2	3.4	32.9	40.8	9.3	33.7
20-SHOT	39.7	0.0	25.0	39.3	0.0	31.5	42.9	2.3	33.7
GPT-J 6B									
5-SHOT	51.6	14.9	38.0	51.3	6.9	42.5	55.8	7.0	44.7
10-SHOT	57.9	9.5	40.0	49.6	3.4	40.4	53.7	9.3	43.7
T5									
SMALL	61.1	32.4	50.5	71.8	10.3	59.6	77.6	30.2	66.8
BASE	68.3	48.6	61.0	65.0	10.3	54.1	84.4	34.9	73.2
LARGE	74.6	44.6	63.5	76.9	24.1	66.4	84.4	51.2	76.8

Experiment 2: Grammar Coverage ---

Goal: Is the grammar expressive enough to capture all XAI questions?

Method

- Use a curated **XAI question bank** (31 questions, informed by design expert interviews)
- Manually review if grammar operations can answer each question

Result

30/31 questions can be answered by the grammar.
The remaining question was deemed out of scope.

(Liao et al. 2020)

Experiment 3: User Study ---

- **45 healthcare workers** + 12 ML grad students
- Diabetes dataset
- Answer **10 XAI multiple-choice questions**, divided into 2 blocks of 5: One block was answered using TalkToModel, and the other using **explainerdashboard**
- Metrics
 - ▶ Ease of use, confidence, speed, and likability (subjective)
 - ▶ Completion rate, Accuracy (objective)

Example: “Is glucose more important than age for the model’s predictions for data point 49?”



explainerdashboard vs TalkToModel Interface ---

Model Explainer

Positive class:

Survived



Feature Importances

Classification Stats

Individual Predictions

What if...

Feature Dependence

Feature Interactions

Decision Trees

Feature Importances

Which features had the biggest impact?

Importances type:

- ☐ Permutation Importances
☒ SHAP values

Depth:

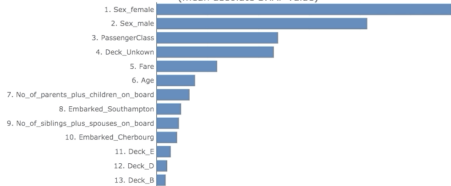
Select...

Grouping:



Group cats

Average impact on predicted Survival
(mean absolute SHAP value)



how likely are patients older than forty to have diabetes?

Over 85 cases where **age greater than 40.0** in the data, the model predicts:

- **unlikely to have diabetes**, 44.706% of the time
- **likely to have diabetes**, 55.294% of the time

why?

For instances with **age greater than 40.0** predicted likely to have diabetes:

- **glucose** is the **most important** feature and has a **positive** influence on the predictions
- **bmi** is the **second** most important feature and has a **positive** influence on the predictions
- **age** is the **third** most important feature and has a **positive** influence on the predictions

For instances with **age greater than 40.0** predicted unlikely to have diabetes:

Enter your command! Use the ↑ arrow and ↓ arrow to cycle previous command Send

Help me generate a question about...

Pins

Pin messages and they will appear here!

There are **85** items in the data where **age greater than 40.0**.
Let me know if you want to see their ids.

For **all** the instances in the data, the statistics of **age** in the dataset are:
mean: 34.192
one std: 12.146
min: 21
max: 81

Experiment 3: Results ---

TalkToModel vs. explainerdashboard

- **Over 90% accurate** vs. $\sim 60\%$ with explainerdashboard
- Answered questions in **half the time**
- **Consistently preferred** on ease of use, confidence, speed, and likability

% Questions Completed

Healthcare workers: 86.2% vs. 74.7%

ML grads: 93.9% vs. 73.8%

% Accuracy on Completed

Healthcare workers: 91.8% vs. 66.1%

ML grads: 100% vs. 62.5%

TalkToModel Conclusions ---

Strengths

- **Elegant UI**: natural language makes model interpretation accessible
- **Highly extensible**: handles a variety of XAI methods and problem domains
- **Reasonably accurate**: fine-tuned T5 interprets user intent well
- **Low barrier**: only need your own dataset to deploy

Limitations

- Not tested in **real-world settings**
- No user **flexibility** in explanation selection
- No guarantees on **data quality** for fine-tuning
- No **domain knowledge** beyond the grammar
- Low accuracy on **compositional split**

Discussion Questions: Is the TalkToModel LLM itself interpretable?

Thank You!



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